

Embeddings

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1– Embeddings

Embeddings are a way to represent data—such as words, sentences, images, or even entire documents—as vectors (lists of numbers) in a continuous vector space. These vectors capture the semantic meaning or features of the data, allowing machines to process, compare, and reason about it more effectively.

Data Type	Embedding Type	Example Model
Word	Word Embedding	Word2Vec, GloVe
Sentence	Sentence Embedding	BERT, SBERT, BGE-M3
Image	Image Embedding	ResNet, CLIP
Graph	Graph Embedding	Node2Vec, GraphSAGE
Audio	Audio Embedding	Wav2Vec, Whisper

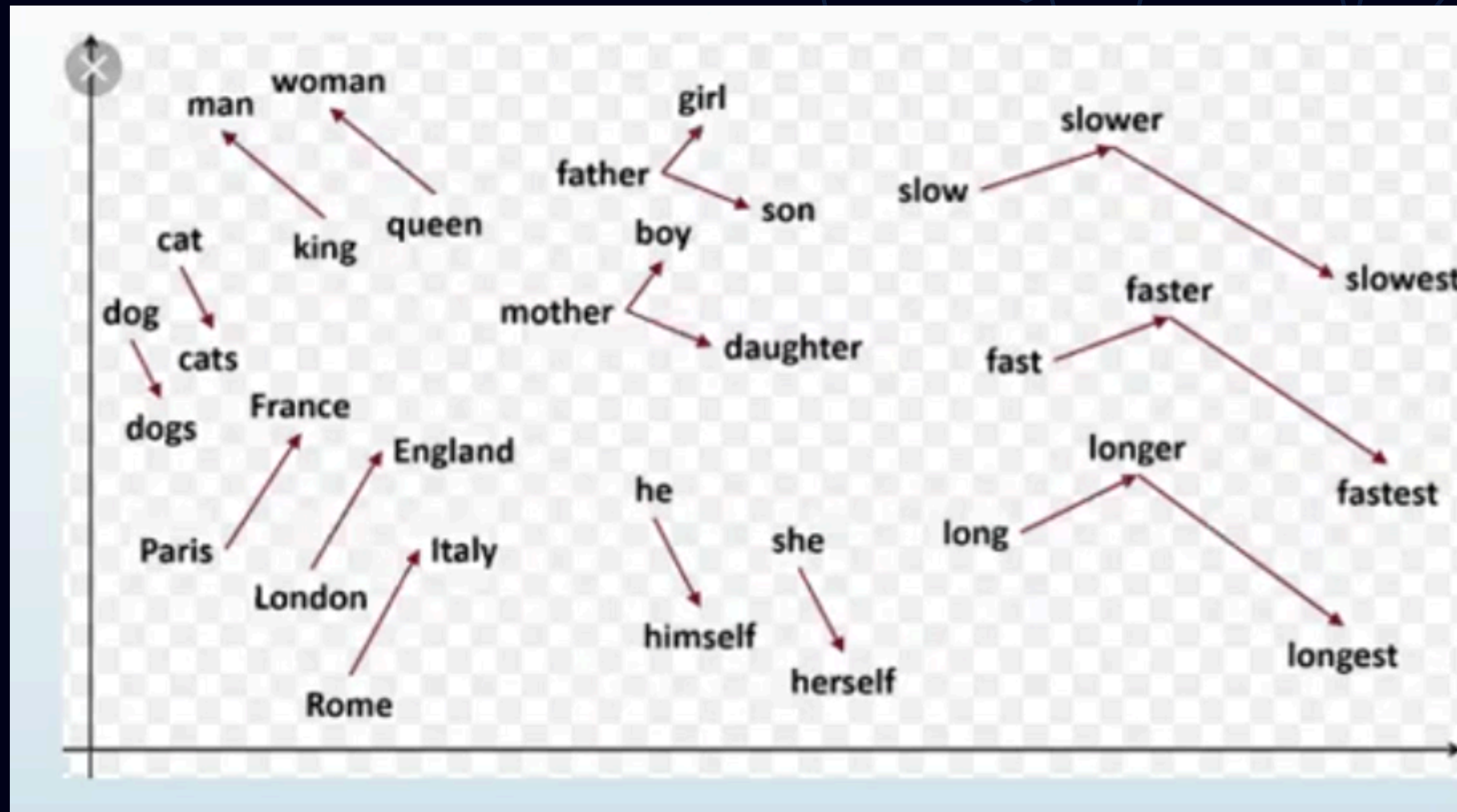
Bag Of Words and TF-IDF Problems

- Both BOW and TF-IDF approach semantic information is not stored. TF-IDF gives importance to uncommon words.
- There is definitely chance of over fitting.

Solution – Word2Vec

- In this specific model, each word is basically represented as a vector of 32 or more dimension instead of a single number
- Here the semantic information and relation between different words is also preserved

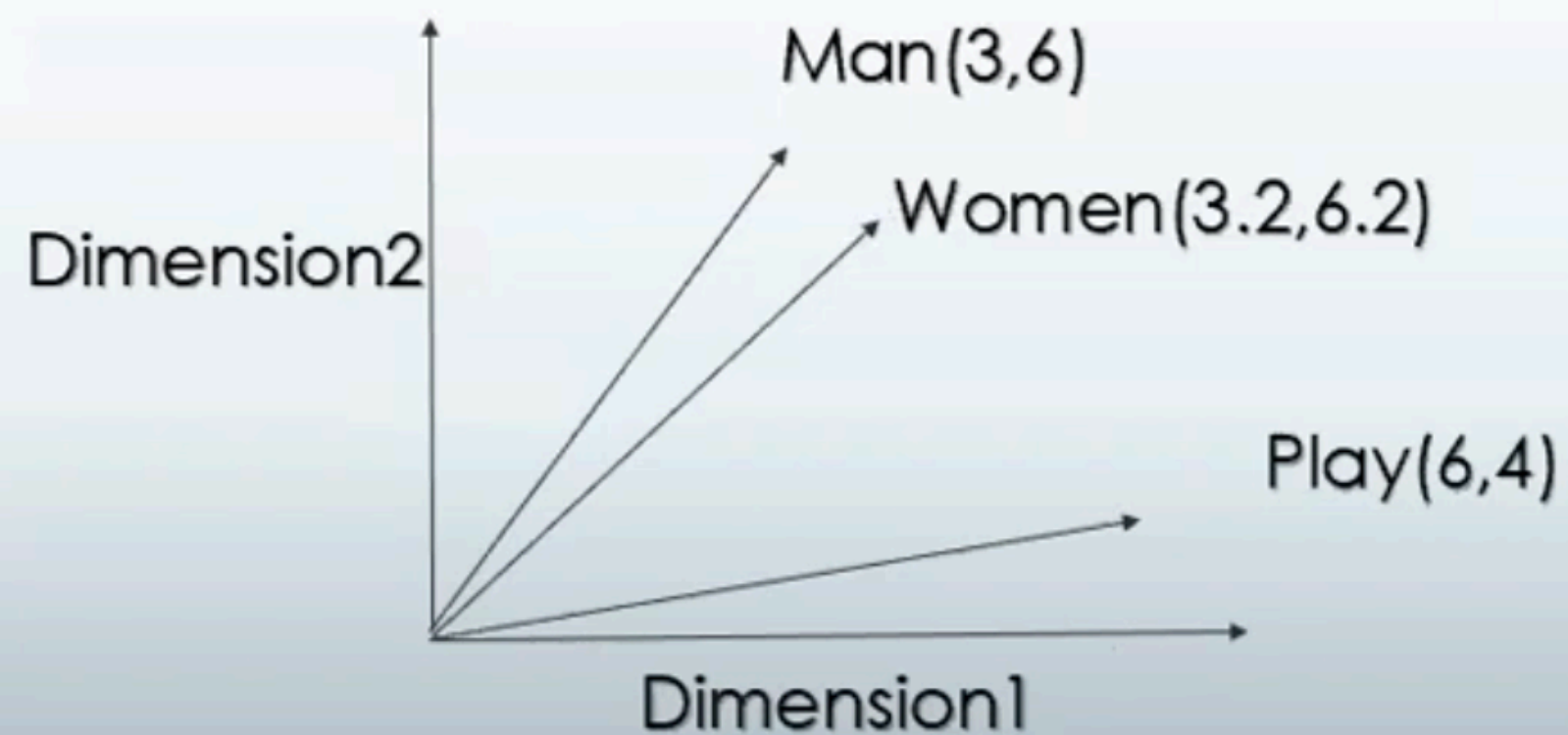
2- word2vec



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Visual Representation- Word2Vec

King - Man + Woman = Queen



Steps to Create Word2vec

- Tokenization of the sentences
- Create Histograms
- Take most frequent words
- Create a matrix with all the unique words. It also represent the occurrence relation between the words

3- Embedding creating

- CBOW
- NEXT WORD
 - UNI-GRAM
 - BI-GRAM
 - N-GRAM

4-average word2vec

You average the Word2Vec vectors of all the individual words in a sentence or document to get a single fixed-size vector that represents the whole sentence or document.